Learnability-Gated Portfolio Expansion

A practical portfolio-manager algorithm implied by the law of portfolio learnability

Keep a broad but admissible opportunity library. Do not rank signals one by one and do not use a binary admission gate. First remove what the current portfolio already spans; then shrink the residual managed-payoff directions according to how strongly the available return history identifies them.

1. One algorithm

Let **B** be the T x p payoff matrix of directions already in the portfolio and **X** a T x q proposed extension. Estimate

```
min_(a,b) ||1 - B a - X b||^2 / T + lambda ||b||^2
```

With P_B=B(B'B)^(-1)B', define H=(I-P_B)X and Sigma_perp=H'H/T. Profiling the baseline gives

```
b_hat=(Sigma_perp+lambda I)^(-1)H'1/T
a_hat=a_hat_0-(B'B)^(-1)B'X b_hat
```

The second equation is not cosmetic: a new signal changes existing positions whenever part of the extension is hedgeable by the baseline.

2. The learnability gate is spectral

Diagonalize Sigma_perp=U diag(mu_1,...,mu_q)U'. Ridge activates each residual direction by

```
g_j(lambda)=mu_j/(mu_j+lambda)
```

Strong directions receive g_j near one; weak directions receive g_j near zero. The incremental effective complexity is

```
C_new(lambda)=sum_j g_j(lambda)=Tr[Sigma_perp(Sigma_perp+lambda I)^(-1)]
```

Thus a library can contain many raw signals while only a finite amount of new portfolio structure becomes statistically active. This is continuous admission, direction by direction - not feature-by-feature acceptance.

3. Monthly operating rule

1. Freeze a curated information set using only data available at the rebalance date.

2. Estimate the current baseline portfolio.

3. Residualize the candidate extension: $H=(I-P_B)X$.

4. Compute the residual managed-payoff spectrum.

5. Choose lambda on a chronological validation block preceding evaluation.

6. Re-estimate both new and baseline positions using b_hat and hedge-adjusted a_hat.

7. Report weights plus lambda, C_new, spectral concentration and tail dependence.

> **The question is not "which feature has the highest Sharpe?" It is "what new portfolio direction remains after the current opportunity set is removed, and how much of it can the present history support?"**

4. What changes relative to feature selection

Conventional	Learnability-based
Rank variables	Preserve a rich admissible library
Keep top features	Residualize against current portfolio
Fit on survivors	Shrink residual payoff directions
Raw count is prominent	Effective payoff dimension is prominent

The recommendation is not to feed arbitrary noise into a kernel. Basic data-quality and timing filters remain essential. The change is to avoid aggressive economic hard-screening inside an admissible library: several characteristics can encode the same portfolio direction, while a weak standalone signal can add useful residual value in combination.

5. Real-data test: JKP portfolios

The project tests this logic on 57 official U.S. JKP capped-value-weighted characteristic-managed portfolio returns. The secondary market-inclusive baseline contains market, size, book-to-market and momentum. Tuning and confirmation precede each outer year in 2005-2025.

T months	Baseline	+ Value	+ Invest.	+ All
120	0.442	0.914	0.688	0.981
240	0.333	1.033	0.603	1.068

At T=120, the all-extension Sharpe increment is 0.540; the reported simultaneous 95% interval for this four-extension diagnostic is [0.053, 1.027]. Additional value directions contribute beyond book-to-market in this comparison.

6. Why the hard gate is not the algorithm

The earlier experimental admission rule required a positive simultaneous lower bound before allowing an extension. With the original three-tilt baseline and T=120, it produced Sharpe 0.208 versus -0.004 for the baseline and 0.691 for the continuously shrunk all-extension policy; it admitted extensions in only 5 of 21 years. The evidence favors continuous spectral shrinkage rather than equating uncertainty with exclusion.

7. Portfolio-manager output

Output	Interpretation
Selected lambda	Shrinkage demanded by chronological evidence
C_new	Effective number of incremental residual directions
Spectral concentration	Whether new value lies in strong directions or a weak tail
Incremental loss gain	Response-one improvement versus the same estimated baseline
Baseline hedge change	How accepted old positions move after the extension enters

8. Connection to the theory

The main theory defines portfolio complexity through the spectrum of managed payoffs. Under polynomial spectral decay, finite market history supports a finite rate-optimal effective dimension. The algorithm applies that logic incrementally: after removing directions already represented by the current portfolio, lambda controls how deeply the investor is allowed to enter the residual opportunity spectrum.

9. Boundaries

This partial-ridge, finite-rank implementation is a research algorithm motivated by the main theory; it does not automatically inherit every theorem proved for the isotropically penalized infinite-dimensional estimator. The 2005-2025 JKP exercise is exploratory rather than a pristine untouched holdout, and it uses characteristic-managed portfolio histories rather than raw stock-level characteristics.

Bottom line

> **Do not ask whether a new signal should be included. Ask which residual portfolio directions it creates and how much of those directions the available market history has earned the right to activate.**

Source basis: Law of Portfolio Learnability manuscript; Incremental Opportunity Learning study; Jensen, Kelly, and Pedersen (2023), Journal of Finance.